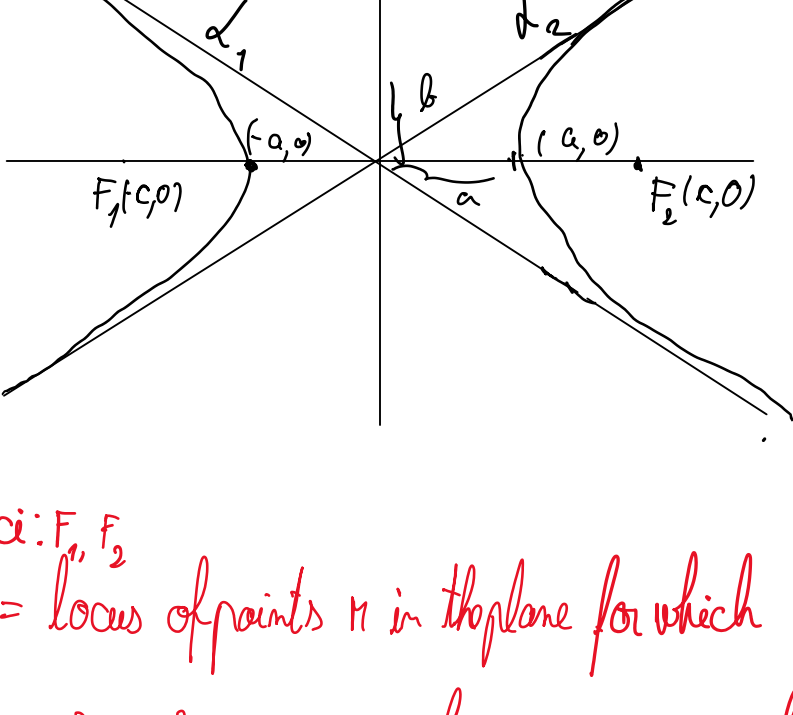


Hyperbolas and parabolas



Foci: F_1, F_2

H = locus of points M in the plane for which $|MF_1 - MF_2| = 2a$

$$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad a, b > 0 \Rightarrow y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$$

$$c = \sqrt{a^2 + b^2}$$

H has 2 asymptotes at $\pm \infty$: $\ell_1: y = -\frac{b}{a}x$
 $\ell_2: y = \frac{b}{a}x$

$$TH(x_0, y_0): \frac{x_0 x}{a^2} - \frac{y_0 y}{b^2} = 1$$

$$\frac{\partial}{\partial x} F(x, y) + \frac{\partial}{\partial y} F(x, y) = 1$$

The tangent to H of a line with slope m : $y = mx \pm \sqrt{a^2 m^2 - b^2}$

1. Determine the intersection points between the line $\ell: 2x - y - 10 = 0$ and the hyperbola $H: \frac{x^2}{25} - \frac{y^2}{9} = 1 = 0$.

$$\begin{cases} 2x - y - 10 = 0 \\ \frac{x^2}{25} - \frac{y^2}{9} = 1 \end{cases} \quad | \cdot 20 \Leftrightarrow \quad x^2 - 4y^2 - 20 = 0$$

$$y = 2(5 - x)$$

$$x^2 - 4 \cdot 4(5 - x)^2 - 20 = 0$$

$$x^2 - 16(25 - 10x + x^2) - 20 = 0$$

$$x^2 - 400 + 160x - 16x^2 - 20 = 0$$

$$-15x^2 + 160x - 420 = 0 \quad | :5$$

$$-3x^2 + 32x - 84 = 0 \quad | : (-3)$$

$$3x^2 - 2 \cdot 16 \cdot (3x) + 28 \cdot 6 - 80x = 0$$

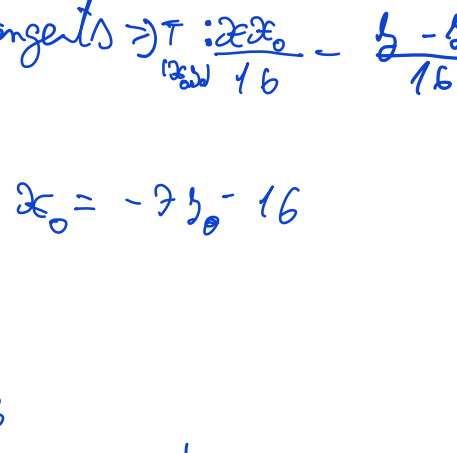
$$(3x - 16)^2 = 50x$$

$$|3x - 16| = 2\sqrt{12.5}$$

$$x = \pm \frac{2\sqrt{12.5}}{3} + 16$$

$$y = 42 \pm 4\frac{\sqrt{12.5}}{3}$$

2. Determine the tangents to the hyperbola $H: \frac{x^2}{16} - \frac{y^2}{8} = 1 = 0$ which are parallel to the line $\ell: 4x + 2y - 5 = 0$.



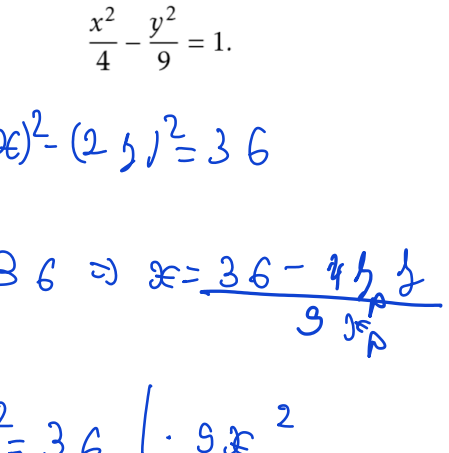
$$\ell: 4x + 2y - 5 = 0 \Rightarrow m = -2$$

$$\frac{x^2}{16} - \frac{y^2}{8} = 1 \Rightarrow a^2 = 16$$

$$b^2 = 8$$

$$y = -2x \pm \sqrt{4 \cdot 16 - 8} = -2x \pm 2\sqrt{14}$$

3. Determine the tangents to the hyperbola $H: x^2 - y^2 = 16$ which contain the point $M(-1, 7)$.



$$H: x^2 - y^2 = 16 \Rightarrow \frac{x^2}{16} - \frac{y^2}{16} = 0$$

Let ℓ_1, ℓ_2 be the tangents $\Rightarrow \frac{x_0 x}{16} - \frac{y_0 y}{16} = 1, \quad P(x_0, y_0) \in H$
 $\forall P \in \ell_1$

$$\begin{cases} -x_0 - 7y_0 = 16 \Leftrightarrow x_0 = -7y_0 - 16 \\ x_0^2 - y_0^2 = 16 \end{cases}$$

$$(-7y_0 - 16)^2 - y_0^2 = 16$$

$$49y_0^2 + 224y_0 + 240 = 0 \quad | : 16$$

$$3y_0^2 + 14y_0 + 15 = 0 \quad | : 3$$

$$(3y_0^2 + 2 \cdot 3y_0 + 2) + 4y_0 + 49 = 0$$

$$(3y_0 + 2)^2 = 4$$

$$y_0 = \begin{cases} -\frac{2}{3} \\ -3 \end{cases}$$

$$x = \begin{cases} -16 - 7 \cdot (-\frac{2}{3}) = -\frac{13}{3} \\ -16 - 7 \cdot (-3) = 5 \end{cases}$$

4. Determine the relations between the coordinates (x_p, y_p) of the point P such that P does not belong to any tangent line to the hyperbola

$$\frac{x^2}{4} - \frac{y^2}{9} = 1.$$

$$\begin{cases} \frac{x^2}{4} - \frac{y^2}{9} = 1 \Leftrightarrow (x^2 - 4y^2) = 36 \\ (3x_p \cdot x - 4y_p \cdot y) = 36 \Rightarrow x = \frac{36 - 4y_p y}{3x_p} \end{cases}$$

$$\left(\frac{36 - 4y_p y}{3x_p}\right)^2 - (2y)^2 = 36 \quad | \cdot 9x_p^2$$

$$(36 - 4y_p y)^2 - 9 \cdot 4 \cdot x_p^2 y^2 = 36 \cdot 9x_p^2$$

$$4^2 (9 - 4y_p y)^2 - 9 \cdot 4 \cdot x_p^2 y^2 = 4 \cdot 9^2 x_p^2$$

$$4(9 - 4y_p y)^2 - 9x_p^2 y^2 = 9^2 x_p^2$$

$$4(81 - 18y_p y + 4y_p^2 y^2) - 9x_p^2 y^2 - 9x_p^2 = 0$$

$$324 - 64y_p y + 44y_p^2 y^2 - 9x_p^2 y^2 - 9x_p^2 = 0$$

$$44y_p^2 - 9x_p^2 y^2 - 64y_p y + 324 - 9x_p^2 = 0$$

5. Find the area of the triangle determined by the asymptotes of the hyperbola $H: \frac{x^2}{9} - \frac{y^2}{9} = 1 = 0$ and the line $\ell: 9x + 2y - 24 = 0$.

$$\ell: 9x + 2y - 24 = 0 \Leftrightarrow y = -\frac{9}{2}x + 12$$

$$H: 9x^2 - 4y^2 - 36 = 0$$

$$\ell_1: y = \frac{3}{2}x$$

$$\ell_2: y = -\frac{3}{2}x$$

$$\ell_1 \cap \ell_2 \begin{cases} y = \frac{3}{2}x \\ y = -\frac{3}{2}x + 12 \end{cases} \Rightarrow \begin{cases} -\frac{3}{2}x = -\frac{3}{2}x + 12 \\ y = -\frac{3}{2}x \end{cases} \Leftrightarrow \begin{cases} x = 4 \\ y = -6 \end{cases}$$

$$\ell_1 \cap \ell_2 \begin{cases} y = \frac{3}{2}x \\ y = \frac{3}{2}x \end{cases} \Leftrightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$$\ell_2 \cap \ell \begin{cases} y = \frac{3}{2}x \\ y = -\frac{9}{2}x + 12 \end{cases} \Leftrightarrow \begin{cases} x = 2 \\ y = 3 \end{cases}$$

$$\Delta = \frac{1}{2} \begin{vmatrix} 2 & 3 & 1 \\ 0 & 0 & 1 \\ 4 & -6 & 1 \end{vmatrix} = \frac{1}{2} (12 + 12) = 12$$

6. Find an equation for the tangent lines to:

a) the hyperbola $H: \frac{x^2}{25} - \frac{y^2}{9} = 1 = 0$, orthogonal to the line $\ell: 4x + 3y - 7 = 0$;

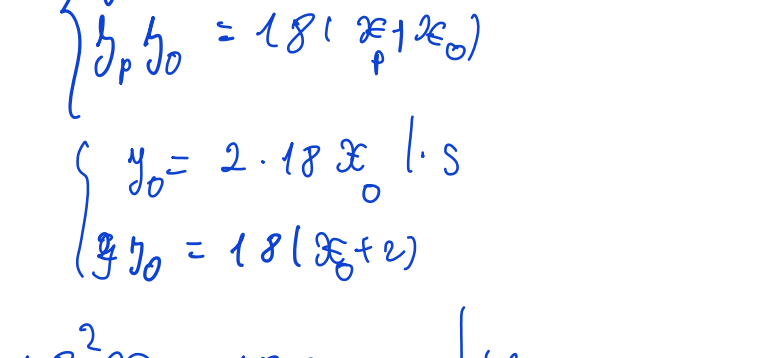
b) the parabola $P: y^2 - 8x = 0$, parallel to $\ell: 2x + 2y - 3 = 0$.

$$a) \ell: 4x + 3y - 7 = 0 \Rightarrow m = -\frac{4}{3}$$

$$\ell_1 \perp \ell \Rightarrow m_{\ell_1} \cdot m_{\ell} = -1 \Rightarrow m_{\ell_1} = \frac{3}{4}$$

$$y = mx \pm \sqrt{a^2 m^2 - b^2} = \frac{3}{4}x \pm \sqrt{25 \cdot \frac{9}{16} - 9} = \frac{3}{4}x \pm \sqrt{\frac{25}{4}} = \frac{3}{4}x \pm \frac{5}{2}$$

b) Parabolas



$$P: y^2 = 2px \Leftrightarrow x = \frac{y^2}{2p}$$

P : locus of points M in the plane for which $MF = \text{dist}(M, d)$, F focus, d line called the directrix

$$TH(x_0, y_0): y_0 = p(x + x_0)$$

The tangent to H of a line with slope m : $y = mx + \frac{p}{2m}$

$$b) y^2 - 8x = 0 \Leftrightarrow y^2 = 2 \cdot 4 \cdot x \Rightarrow p = 4$$

$$TH \Rightarrow m_1 = m_2 \Rightarrow m_1 = -1$$

$$m_2 = -\frac{2}{1} = -1$$

$$TH: y = -x + \frac{4}{(-1)} = -x - 2$$

7. Find an equation for the tangent lines to:

a) the hyperbola $H: \frac{x^2}{9} - \frac{y^2}{9} = 1 = 0$, passing through $P(1, -5)$;

b) the parabola $P: y^2 - 36x = 0$, passing through $P(2, 9)$.

$$a) \frac{x^2}{9} - \frac{y^2}{9} = 1 = 0 \Rightarrow \begin{cases} 5x_0^2 - 3y_0^2 = 15 \\ \frac{x_0}{3} - \frac{y_0}{9} = 1 \end{cases} \Leftrightarrow \begin{cases} 5x_0^2 - 3y_0^2 = 15 \\ x_0 + 3y_0 = 3 \end{cases} \Leftrightarrow \begin{cases} 5x_0^2 - 3y_0^2 = 15 \\ x_0 = 3(1 - y_0) \end{cases}$$

$$5 \cdot 9(1 - y_0)^2 - 3y_0^2 = 15 \quad | : 3$$

$$15(1 - 2y_0 + y_0^2) - 3y_0^2 = 15$$

$$15 - 30y_0 + 15y_0^2 - 3y_0^2 = 15$$

$$14y_0^2 - 30y_0 = 0 \quad | : 2$$

$$7y_0^2 - 15y_0 = 0 \Rightarrow \begin{cases} y_0 = 0 \Rightarrow x_0 = 3 \\ y_0 = \frac{15}{7} \Rightarrow x_0 = -\frac{24}{7} \end{cases}$$

$$b) \begin{cases} y_0^2 - 36x_0 = 0 \Rightarrow y_0^2 = 2 \cdot 18x_0 \\ y_0 y_0 = 18(1 + x_0) \end{cases}$$

$$\begin{cases} y_0 = 2 \cdot 18x_0 \quad | : 5 \\ 18y_0 = 18(x_0 + 2) \end{cases}$$

$$18^2 x_0 = 18(x_0 + 2) \quad | : 18$$

$$18x_0 = x_0 + 2$$

$$x_0 = \frac{2}{17} \quad \rightarrow \quad \frac{2 \cdot 2}{17} y = 18(x + \frac{2}{17})$$

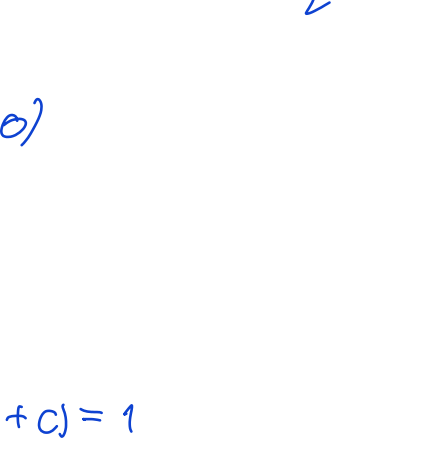
$$y_0 = \frac{36}{17} \quad \rightarrow \quad 2y = 18(17x + 2)$$

8. Consider the hyperbola $H: x^2 - y^2 = 1 = 0$ with focal points F_1 and F_2 . Find the points M situated on the hyperbola such that

a) The angle $\angle F_1 M F_2$ is right;

b) The angle $\angle F_1 M F_2$ is 60° ;

c) The angle $\angle F_1 M F_2$ is θ .



$$\frac{x^2}{1} - \frac{y^2}{1} = 1 = 0 \Rightarrow \begin{cases} a = 1 \\ b = 1 \\ c = \sqrt{1^2 + 1^2} = \sqrt{2} \end{cases}$$

$$F_1(-\sqrt{2}, 0), \quad F_2(\sqrt{2}, 0)$$

$$M(x, y)$$

$$M_{F_1}(-\sqrt{2} - x, -y)$$

$$M_{F_2}(\sqrt{2} - x, -y)$$

$$M_{F_1} \cdot M_{F_2} = \|M_{F_1}\| \cdot \|M_{F_2}\| \cdot \cos \theta$$

$$F_1 \cdot F_2 = \theta$$

$$(\sqrt{2} - x, -y) \cdot (\sqrt{2} - x, -y) = \sqrt{(x - \sqrt{2})^2 + y^2} \cdot \sqrt{(\sqrt{2} + x)^2 + y^2} \cdot \cos \theta$$

$$-(\sqrt{2} - x)(\sqrt{2} + x) + y^2 = \sqrt{(x - \sqrt{2})^2 + y^2} \cdot \sqrt{(\sqrt{2} + x)^2 + y^2} \cdot \cos \theta$$

$$\frac{x^2 + y^2 - 5}{\sqrt{(x - \sqrt{2})^2 + y^2} \cdot \sqrt{(x + \sqrt{2})^2 + y^2}} = \cos \theta$$

$$y^2 = 4x^2 - 1$$

$$x^2 + y^2 = 5 = x^2 + 4x^2 - 1 = 5x^2 - 6$$

$$(x - \sqrt{2})^2 + y^2 = x^2 - 2\sqrt{2}x + 5 + 4x^2 - 1 = 5x^2 - 2\sqrt{2}x - 15$$

$$(x + \sqrt{2})^2 + y^2 = 5x^2 + 2\sqrt{2}x + 5$$

$$((x - \sqrt{2})^2 + y^2)((x + \sqrt{2})^2 + y^2) = 25(x^2 + 1) - 20x^2 = 25(x^4 + 2x^2 + 1) - 20x^2 = 25x^4 + 30x^2 + 25$$

$$\cos \theta = \frac{5x^2 - 6}{\sqrt{25x^4 + 30x^2 + 25}}$$

$$t = x^2$$

$$\cos \theta = \frac{\sqrt{t} - 6}{\sqrt{25t^2 + 30t + 25}}$$

10. Consider the hyperbola $H: x^2 - 2y^2 = 1$. Determine the geometric locus described by the midpoints of the chords of H which are parallel to the line $2x - y = 0$.

$$x^2 - 2y^2 = 1$$

$$\frac{x^2}{1} - \frac{y^2}{\frac{1}{2}} = 1 \Rightarrow c = \sqrt{a^2 + b^2} = \frac{\sqrt{5}}{2}$$

$$F_1(-\frac{\sqrt{5}}{2}, 0), \quad F_2(\frac{\sqrt{5}}{2}, 0)$$

$$\begin{cases} x^2 - 2y^2 = 1 \\ y = 2x + c \end{cases}$$

$$x^2 - 2(4x^2 + 4xc + c^2) = 1$$

$$x^2 - 8x^2 - 8xc - 2c^2 - 1 = 0$$

$$-7x^2 - 8xc - 2c^2 - 1 = 0$$

$$x_1 + x_2 = -\frac{8c}{7}$$

$$x_M = \frac{x_1 + x_2}{2} = -\frac{4}{7}c$$

$$y_M = -\frac{8x}{7} + c = -\frac{4}{7}c$$

$$M(-\frac{4}{7}c, -\frac{4}{7}c) \Rightarrow V = \langle 1, 1 \rangle$$